

RFM6505W LoRaWan Node Module User Guide

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1. Module Features

6505 LoraWan node module is a wireless serial port module based on LoRaWan ClassA protocol. It has the advantages of selectable output power, high sensitivity, long transmission distance, and small current at low power consumption.

Simple user interface and convenient application operation. The user MCU only needs to connect to the module through the serial port, and send data to the module through the serial port when sending data packets, and the module will package and transmit the user data packets to the gateway in the LoRaWan protocol and upload them to the background. Receiving downlink data packets according to standard protocols.

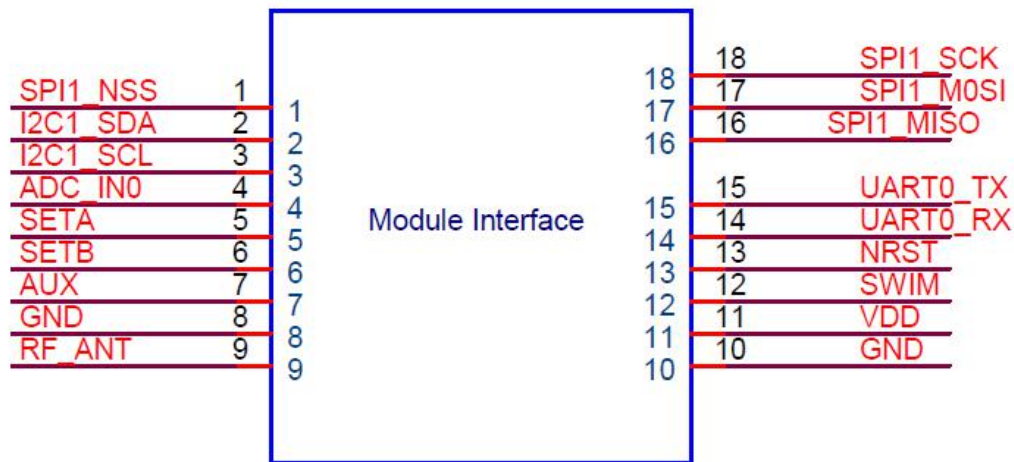
The user can control the working mode of the module through the mode control pins SETA and SETB. The module can work in: 1. Regular data transceiver mode; 2. Parameter configuration mode; 3. Sleep mode. (Refer to Section 2 for specific operations).

Provide a simple AT command set for setting various parameters of the module. Such as: network access activation mode, network access triples, port number, rate adaptive ADR selection, transmission power, etc. (Refer to Section 3 for specific application of instruction format).

This module mainly has four frequency bands according to LORAWAN regional documents: EU433MHZ ,CN470MHZ, EU 863-870MHz, US 902-928MHz. Different regional regulations use different frequency bands. There are four types of 6505 modules: RFM6505-433S2, RFM6505-470S2, RFM6505-868S2 and RFM6505-915S2. Each of them conforms to LoRaWan's corresponding regional specification documents such as: "LoRaWAN 1.0.2 Regional Parameters". But it should be noted that: due to the limited RAM space resources of the MCU inside the module chip, the maximum value of the 915MHZ frequency band data packet cannot reach the 242 bytes specified in the specification document, and the maximum value is only 200 bytes.

2. User Interfaces

2.1 Overall Interface to User



2.2 Minimal User Interface

The minimum interfaces of the module to the user are: SETA、SETB、TXD、RXD、AUX. The functions of these IO ports are as follows:

Module Pin	IO Attributes
SETA	Input
SETB	Input
TXD	Output
RXD	Input
AUX	Output

The control logic of the module working mode selection pins SETA and SETB is as follows:

Pin SETA State	Pin SETB State	Module State
H	X (High and low does not matter)	Low Power State
L	L	Parameter Configuration Status
L	H	Data Receiving & Sending Status

✚ SETA working mode and low power consumption mode selection pin.

Module SETA pin enables internal pull-up by default. It defaults to high without connecting any external devices. When SETA is high, the module enters low power consumption mode and stays dormant. In the sleep state, the module does not detect the state of SETB in the mode. And in this mode, it does not receive any data from UART or output any data. After SETA is pulled down, the module exits the low-power mode and enters the working mode. There are two working modes: 1.

Data transceiver mode, 2. Parameter configuration mode. Which mode to enter depends on the state of SETB. It should be noted that after the module is reset, it has been trying to access the network before it has been connected to the network. Only after the network access is OK, SETA can go to sleep after returning to high level.

✚ SETB data transceiver mode and parameter configuration mode selection pin

SETB needs to be pulled down before SETB works. Module SETB pin enables internal pull-up by default. It defaults to high without connecting any external devices. When SETB is high, it is the data sending and receiving mode. After SETB is pulled low, it is the parameter configuration mode. Send AT commands to the module through the serial port to query and modify the parameters.

✚ TXD RXD is the module's external UART data interface

The TXD of the module is connected to the RXD of the user, and the RXD of the module is connected to the TXD of the user. The module receives the user's data packet through RXD, and transmits the data packet by radio frequency. If the module receives data through the radio frequency channel, it will send the data packet to the user through the TXD pin.

✚ AUX is the working status indicator pin of the module

The default output of AUX is high. When the user sends a data packet to the module through the serial port, pulling down the module AUX indicates that the module processes the user data packet. Therefore, when the user needs to continuously send multiple packets of data to the module when operating the module, it is necessary to determine the state of the AUX to avoid packet loss. If AUX is low, you need to wait, if AUX becomes high, you can send data packets.

2.3 Reserved Expansion Interface

Other pins of the module are reserved for subsequent expansion functions. Mainly have ADC, I2C, SPI interface. These interfaces can also be configured as ordinary IO ports. It should be noted that the module requires an external pull-up resistor when using the I2C interface.

3. AT Instruction Set

3.1 AT Command Syntax

The AT command uses a command line based on ASCII code, and the command format is as follows:

The request message format is: AT+<CMD>[OP][para-1,para-2,.....para-n]<\r>

Table1: AT Request Message Format

Area	Description
AT+	Command Message Prefix
CMD	Command String
Op	Instruction Operator. It can be the following: ✓ “=”: Indicates parameter settings. ✓ “?”: Represents the current value of the query parameter. ✓ “”: Indicates that the instruction is executed. ✓ “=? ”: Indicates the parameter of the query setting instruction.
para-1,para-2,.....para-n	Indicates the parameter value to be set, or specifies the parameter to be queried.
\r	Carriage Return Terminator. ASCII code is 0x0D

The response message format is: <\r\n>[+CMD:][para-1,para-2,.....para-n]<\r\n>

Or: <\r\n><STATUS><\r\n>

Or both.

Table2: AT Response Message Format

Area	Description
\n	Newline Character. ASCII code is 0x0A
+CMD	Corresponding Command String
para-1,para-2,.....para-n	Corresponding Parameter String
STATUS	The instruction execution status. It can be the following: ✓ “OK”: Indicates that the instruction was successfully executed ✓ “ERROR”: Indicates that the command execution failed ✓ “+CME ERROR:<err>”: Indicates that the command execution failed and the corresponding error code is returned.

Note:

- ◇: Indicates what must be included
- [: Indicates optional content
- \r: Carriage return terminator, ASCII code is 0x0D
- \n: Newline character, ASCII code is 0x0A.

For example, to query the network activation mode, the sending instruction is:

AT+CJOINMODE?\r

The reply message is:

\r\n++CJOINMODE:0\r\n

\r\nOK\r\n

In the following text, \r\n is hidden for readability.

- Parameter configuration of the serial port: baud rate 115200, data bit 8, stop bit 1, parity bit 0.
- The current command supports echoing, but does not support BackSpace for the time being, and does not support backspace for history commands.

3.2 Command Classification

LoRa's AT commands are classified as follows:

Classification	Description	Remark:
General Command	Manufacturer identification, module identification, version identification, product serial number identification;	General commands
Network Related Parameter Configuration Commands	Band mask, multicast address, same frequency/different frequency, device DevEUI;	Network related parameters config commands
MAC Configuration Commands	Related to MAC commands in LoRaWAN protocol;	MAC config commands

3.2.1 General Command Overview

Command	Description	Method to Realize
AT+CGMR	Read version ID (revision identification)	Optional
AT+CGBR	Set the baud rate of the UART(baud rate on UART interface)	Optional

3.2.2 Overview of Network Related Parameter Configuration Commands

Command	Description	Method to Realize
AT+CJOINMODE	Set read Join mode (OTAA, ABP)	Required
AT+CDEVEUI	Set to read DevEUI (when OTAA is online)	Required
AT+CAPPEUI	Set to read AppEUI (when OTAA is online)	Required
AT+CAPPKEY	Set to read AppKey (when OTAA accesses the network)	Required
AT+CDEVADDR	Set to read DevAddr (when ABP is connected to the network)	Required
AT+CAPPKEY	Set to read AppSKey (when ABP is connected to the network)	Required
AT+CNWKSKEY	Set to read NwkSKey (when ABP is connected to the network)	Required
AT+CFREQBANDMASK	Set the read frequency point mask (FreqBandMask)	Required

3.2.3 Overview of MAC Configuration Commands

Command	Description	Method to Realize
AT+CCONFIRM	Set the type of reading sent message (confirm or unconfirm)	Required
AT+CAPPPOINT	Set read application layer Port	Required
AT+CDATARATE	Set the read data rate	Required
AT+CNBTTRIALS	Set read NbTrans parameters	Required
AT+CRM	Set read report mode	Required
AT+CTXP	Set read transmit power	Required
AT+CADR	Enable or disable ADR	Required
AT+CRXP	Set read receive window parameters	Required
AT+CRX1DELAY	Set the delay of reading TX and RX1	Required
AT+CSAVE	Save configuration	Required
AT+CRESTORE	Restore default configuration	Required

3.3 Detailed Instructions

3.3.1 Read Version ID +CGMR

Command Type	Command Format	Response
Query Command	AT+CGMR?	+CGMR=<revision> OK
Parameter Description	<revision>: Version identification	
Return Value Description		
Example	AT+CGMR? +CGMR=v1.0 OK	
Precautions		

3.3.2 Set the Baud Rate +CGBR

Command Type	Command Format	Response
Query Command	AT+CGBR?	+CGBR=<baud> OK
Set Command	AT+CGBR=<baud>	OK
Parameter Description	<baud>: Serial port baud rate	
Return Value Description		
Example	AT+CGMR? +CGMR=v1.0 OK	
Precautions		

3.3.3 Set Join Method +CJOINMODE

Command Type	Command Format	Response
Test Command	AT+CJOINMODE=?	+CJOINMODE:"mode" OK
Query Command	AT+CJOINMODE?	+CJOINMODE:<mode> OK
Run Command	AT+CJOINMODE=<mode>	OK OR +CME ERROR:<err>
Parameter Description	<mode>: Node Join mode is as follows:	
Return Value Description	0: OTAA 1: ABP <err>: error code.	

Example	AT+CJOINMODE=0 OK
Precautions	OTAA method is adopted by default; If you need to use ABP to access the network, please use this command to set before sending data.

3.3.4 Set DevEUI +CDEVEUI

Command Type	Command Format	Response
Test Command	AT+CDEVEUI=?	+ CDEVEUI =< DevEUI:length is 16>
Query Command	AT+ CDEVEUI?	+ CDEVEUI:<value> OK
Run Command	AT+ CDEVEUI =<value>	OK OR +CME ERROR:<err>
Parameter Description	<value>: Node DevEUI	
Return Value Description	<err>: error code.	
Example	AT+ CDEVEUI =AABBCCDD00112233 OK	
Precautions	Set or read DevEUI, return Y1Y2...Y8 in hexadecimal format, the value is 8 bytes.	

3.3.5 Set AppEUI +CAPPEUI

Command Type	Command Format	Response
Test Command	AT+CAPPEUI=?	+CAPPEUI=<AppEUI:length is 16>
Query Command	AT+CAPPEUI?	+CAPPEUI:<value> OK
Run Command	AT+CAPPEUI=<value>	OK OR +CME ERROR:<err>
Parameter Description	<value>: Node AppEUI	
Return Value Description	<err>: error code.	
Example	AT+CAPPEUI=AABBCCDD00112233 OK	
Precautions	Used in OTAA, set or read AppEUI, return Y1Y2...Y8, hexadecimal format, the value is 8 bytes.	

3.3.6 Set AppKey +CAPPKEY

Command Type	Command Format	Response
Test Command	AT+CAPPKEY=?	+CAPPKEY=<AppKey:length is 32>
Query Command	AT+CAPPKEY?	+CAPPKEY:<value> OK
Run Command	AT+CAPPKEY=<value>	OK OR +CME ERROR:<err>
Parameter Description	<value>: Node AppKey	
Return Value Description	<err>: error code.	
Example	AT+CAPPKEY=AABBCCDD00112233AABBCCDD00112233 OK	
Precautions	Used in OTAA, set or read AppKey, return Y1Y2...Y16, hexadecimal format, the value is 16 bytes.	

3.3.7 Set DevAddr +CDEVADDR

Command Type	Command Format	Response
Test Command	AT+CDEVADDR=?	+CDEVADDR=<DevAddr:length is 8, Device address of ABP mode>
Query Command	AT+CDEVADDR?	+CDEVADDR:<value> OK
Run Command	AT+CDEVADDR=<value>	OK OR +CME ERROR:<err>
Parameter Description	<value>: Node DevAddr	
Return Value Description	<err>: error code.	
Example	AT+CDEVADDR=00112233 OK	
Precautions	Used in ABP, set or read DevAddr, return Y1Y2...Y4, hexadecimal format, the value is 4 bytes.	

3.3.8 Set AppSKey +CAPPSKEY

Command Type	Command Format	Response
Test Command	AT+CAPPSKEY=?	+CAPPSKEY=<AppSKey:length is 32>
Query Command	AT+CAPPSKEY?	+CAPPSKEY:<value> OK

Run Command	AT+CAPPSKEY=<value>	OK OR +CME ERROR:<err>
Parameter Description	<value>: Node AppSKey	
Return Value Description	<err>: error code.	
Example	AT+CAPPSKEY=AABBCCDD00112233AABBCCDD00112233 OK	
Precautions	Use in ABP, set or read AppSKey, return Y1Y2...Y16, hexadecimal format, value 16 bytes.	

3.3.9 Set NwkSKey +CNWKSKEY

Command Type	Command Format	Response
Test Command	AT+CNWKSKEY=?	+CNWKSKEY =<NwkSKey:length is 32>
Query Command	AT+CNWKSKEY?	+CNWKSKEY:<value> OK
Run Command	AT+CNWKSKEY=<value>	OK OR +CME ERROR:<err>
Parameter Description	<value>: Node wkSKey	
Return Value Description	<err>: error code.	
Example	AT+CNWKSKEY=AABBCCDD00112233AABBCCDD00112233 OK	
Precautions	Used in ABP, set or read NwkSKey, return Y1Y2...Y16, hexadecimal format, the value is 16 bytes.	

3.3.10 Set Frequency Band Mask +CFREQBANDMASK

Command Type	Command Format	Response
Test Command	AT+CFREQBANDMASK=?	+CFREQBANDMASK:"mask" OK
Query Command	AT+CFREQBANDMASK?	+CFREQBANDMASK:<mask> OK
Run Command	AT+CFREQBANDMASK=<m ask>	OK OR +CME ERROR:<err>
Parameter Description	<mask>: The frequency point mask that the network may work, 16bit corresponds to 16 frequency groups, see LoRaWAN access specification for details.	
Return Value Description	For example: 0-7 channels, the corresponding mask is 0001, 8-15 channels, the corresponding mask is 0002, and so on.	

	The frequency corresponding to the specific channel needs to check the region protocol. For example, channels 0-7 correspond to the following in CN470: 470.3MHz, 470.5MHz, 470.7MHz, 470.9MHz, 471.1MHz, 471.3MHz, 471.5MHz, 471.7MHz. <err>: error code
Example	AT+CFREQBANDMASK=0001 OK
Precautions	It needs to be set before Join.

3.3.11 Set the Upstream Transmission Type +CCONFIRM

Command Type	Command Format	Response
Test Command	AT+CCONFIRM=?	+CCONFIRM:"value" OK
Query Command	AT+CCONFIRM?	+CCONFIRM:<value> OK
Run Command	AT+CCONFIRM=<value>	OK OR +CME ERROR:<err>
Parameter Description	<value>: as follows:	
Return Value Description	0:Unconfirmed Up Message 1: Confirmed Up Message <err>: error code	
Example	AT+CCONFIRM=1 OK	
Precautions	It needs to be set before sending data.	

3.3.12 Set the Upstream Data Port Number +CAPPPORT

Command Type	Command Format	Response
Test Command	AT+CAPPPORT=?	+CAPPPORT:"value" OK
Query Command	AT+CAPPPORT?	+CAPPPORT:<value> OK
Run Command	AT+CAPPPORT=<value>	OK OR +CME ERROR:<err>
Parameter Description	<value>: as follows:	
Return Value Description	The port used, the data format is decimal, the factory value is 10 Value Range: 1~223; Note 1: Port:0x00 is the MAC command of LoRaWAN <err>: error code.	

Example	AT+CAPPPOINT=10 OK
Precautions	It needs to be set before sending data.

3.3.13 Set Communication Rate +CDATARATE

Command Type	Command Format	Response
Test Command	AT+CDATARATE=?	+CDATARATE:"value" OK
Query Command	AT+CDATARATE?	+CDATARATE:<value> OK
Run Command	AT+CDATARATE=<value>	OK OR +CME ERROR:<err>
Parameter Description	<value>: as follows:	
Return Value Description	Rate value, value range: 0 - SF12, BW125 1 - SF11, BW125 2 - SF10, BW125 3 - SF9, BW125 4 - SF8, BW125 5 - SF7, BW125 <err>: error code.	
Example	AT+CDATARATE=1 OK	
Precautions	It needs to be set before sending data.	

3.3.14 Set Sending Times +CNBTRIALS

Command Type	Command Format	Response
Test Command	AT+CNBTRIALS=?	+CNBTRIALS: "MType", "value" OK
Query Command	AT+CNBTRIALS?	+CNBTRIALS:<MType>,<value> OK
Run Command	AT+CNBTRIALS=<MType>,<value>	OK OR +CME ERROR:<err>
Parameter Description	<MType>:0:unconfirm package, 1:confirm package	
Return Value Description	<value>:Is the maximum number of transmissions, the value range: 1~15. The default value is detailed in the access specification. <err>: error code.	

Example	AT+CMBTRIALS=1,2 OK
Precautions	Need to set before sending data.

3.3.15 Set Escalation Mode +CRM

Command Type	Command Format	Response																					
Test Command	AT+CRM=?	+CRM:"reportMode","reportInterval" OK																					
Query Command	AT+CRM?	+CRM:<reportMode>,[reportInterval] OK																					
Run Command	AT+CRM=<report Mode>, [reportInterval]	OK OR +CME ERROR:<err>																					
Parameter Description	This command is mainly used for testing purposes.																						
Return Value Description	<reportMode>: 0- Aperiodic report data; 1- Report data periodically; <reportInterval>: Parameters are only available when reporting data periodically. The interval of reporting data periodically, in s. For different DRs, the minimum allowable period is different, and the period level definition is adopted, as shown in the following table. <table border="1"> <thead> <tr> <th>Rate \period (s)\level</th><th>LV1</th><th>LV2</th></tr> </thead> <tbody> <tr> <td>DR0</td><td>150</td><td>300</td></tr> <tr> <td>DR1</td><td>75</td><td>150</td></tr> <tr> <td>DR2</td><td>35</td><td>70</td></tr> <tr> <td>DR3</td><td>15</td><td>30</td></tr> <tr> <td>DR4</td><td>10</td><td>20</td></tr> <tr> <td>DR5</td><td>5</td><td>10</td></tr> </tbody> </table> <err>: error code.		Rate \period (s)\level	LV1	LV2	DR0	150	300	DR1	75	150	DR2	35	70	DR3	15	30	DR4	10	20	DR5	5	10
Rate \period (s)\level	LV1	LV2																					
DR0	150	300																					
DR1	75	150																					
DR2	35	70																					
DR3	15	30																					
DR4	10	20																					
DR5	5	10																					
Example	AT+CRM=1,2 OK																						
Precautions	Need to set before sending data.																						

3.3.16 Set Transmit Power +CTXP

Command Type	Command Format	Response
Test Command	AT+CTXP=?	+CTXP:"value" OK
Query Command	AT+CTXP?	+CTXP:<value> OK
Run Command	AT+CTXP=<value>	OK OR +CME ERROR:<err>
Parameter Description	<value>: Is the transmission power, the factory value is 0,	
Return Value Description	The actual value range is related to the specific product model, the value range of CN470: 0 - 19dBm 1 - 17dBm 2 - 15dBm 3 - 13dBm 4 - 11dBm 5 - 9dBm 6 - 7dBm 7 - 5dBm <err>: error code.	
Example	AT+CTXP=1 OK	
Precautions	It needs to be set before sending data.	

3.3.17 Enable ADR +CADR

Command Type	Command Format	Response
Test Command	AT+CADR=?	+CADR:"value" OK
Query Command	AT+CADR?	+CADR:<value> OK
Run Command	AT+CADR=<value>	OK OR +CME ERROR:<err>
Parameter Description	<value>: as follows:	
Return Value Description	ADR enable control, factory value is 1 0 - ADR Disable 1 - ADR Enable <err>: error code.	
Example	AT+CADR=1	

	OK
Precautions	It needs to be set before sending data. ADR is enabled by default.

3.3.18 Set the Receiving Window Parameters +CRXP

Command Type	Command Format	Response
Test Command	AT+CRXP=?	+CRXP:"RX1DRoffest","RX2DataRate","RX2Frequency" OK
Query Command	AT+CRXP?	+CRXP:<RX1DRoffest>,<RX2DataRate>,<RX2Frequency> OK
Run Command	AT+CRXP=<RX1DRoffest>,<RX2DataRate>,<RX2Frequency>	OK OR +CME ERROR:<err>
Parameter Description	<RX1DRoffest>,<RX2DataRate>,<RX2Frequency>See details in LoRaWAN protocol <err>: error code.	
Return Value Description		
Example	AT+CRXP=1,1,471000000 OK	
Precautions	It needs to be set before sending data. Do not set the default value.	

3.3.19 Set Sending & Receiving Delay +CRX1DELAY

Command Type	Command Format	Response
Test Command	AT+CRX1DELAY=?	+CRX1DELAY:"Delay" OK
Query Command	AT+CRX1DELAY?	+CRX1DELAY:<Delay> OK
Run Command	AT+CRX1DELAY=<Delay>	OK OR +CME ERROR:<err>
Parameter Description	Delay: The time to open the RX1 window after sending, unit: s; <err>: error code.	
Return Value Description		
Example	CRX1DELAY=2 OK	
Precautions	Set the time to open the RX1 window after sending, set before sending data. Default value of protocol when not set	

3.3.20 Save MAC parameter settings +CSAVE

Command Type	Command Format	Response
Test Command	AT+CSAVE =?	+CSAVE OK
Query Command	AT+CSAVE	OK OR +CME ERROR:<err>
Run Command	This command saves the configuration parameters to EERPOM/FLASH, and the module will use the new MAC configuration parameters for network initialization and operation after restart <err>: error code.	
Parameter Description		
Return Value Description	AT+CSAVE OK	
Precautions	Need to save before sending data.	

3.3.21 Restore MAC default parameters +CRESTORE

Command Type	Command Format	Response
Test Command	AT+CRESTORE=?	+CRESTORE OK
Query Command	AT+CRESTORE	OK OR +CME ERROR:<err>
Run Command	This command restores MAC default configuration parameters to EERPOM/FLASH parameters for network initialization and operation <err>: error code.	
Parameter Description		
Return Value Description	AT+CRESTORE OK	
Precautions	Need to save before sending data	